

Insider Trading in the US Congress

2739941, 2852988, 2869153, 2841038, 2766610, Joshua Azadi and 2865081

2025-06-19

Set-up your environment

Title Page

Dewin Erkamps, Alec van der Heijde, Berend Hol, Dangy Hlynisdóttir, Giovanni Vittorio Boba, Joshua Azadi and Mark de Jongh

Group 4 (4)

C. Schouwenaar (Chantal)

Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

Public confidence in members of congress' ability to act in the best interests of the country, uninfluenced by personal financial incentives, is essential to maintaining the integrity of democratic institutions. However, recent studies have brought up worries about possible conflicts of interest resulting from American congress members' stock market dealings. A significant study published by Abdurakhmonov et al. discovered that U.S. senators' stock purchases frequently resulted in abnormal positive returns, especially when the senators had jurisdiction over the companies through committee assignments or when the companies had made political connections, such as through lobbying or political donations (Abdurakhmonov et al., 2022). The Stop Trading on Congressional Knowledge (STOCK) Act of 2012 sought to address insider trading by forcing members of Congress to make prompt financial transaction disclosures, but enforcement has been difficult. For example, in the beginning of the COVID-19 pandemic in 2020, a number of senators came under fire for stock market activity that occurred after they had received classified briefings. The current regulatory system may have flaws, as evidenced by the fact that no charges have been brought despite the start of the current framework (Campaign Legal Center, n.d.). In response, legislative attempts have continued to these persistent concerns. By expressly defining and prohibiting such behavior, the Insider Trading Prohibition Act (H.R. 2655), which the House approved in 2021, seeks to strengthen prohibitions against trading on material, nonpublic information. The goal of this programming project is to evaluate publicly accessible financial disclosure data in order to find patterns that might point to insider trading by members of congress. By developing tools that analyze congressional financial disclosures and highlight possible illegal transactions, this programming project seeks to help correct that disparity. This initiative aims to increase the transparency of these financial activities by employing data science and programming methods to link transactions to market happenings, legislative changes, and committee assignments.

Part 2 - Data Sourcing

2.1 Load in the data

```
congresstrades <- read.csv("data/Copyofcongress-trading-all.csv")
senators_full_data<-read.csv("data/Copyofcongress-trading-all.csv")
SP_500_annual_returns <- read_csv("data/sp-500-historical-annual-returns.csv" , skip = 15)
##skip metadata
SP_500_annual_returns <- as.data.frame(SP_500_annual_returns) ##set data to class = dataframe
sp_500_monthly_full_data <- read.csv("data/sp500_monthly.csv")
```

2.2 Provide a short summary of the dataset(s)

```
head(as_tibble( senators_full_data))
```

```
## # A tibble: 6 x 19
##   Ticker TickerType Company Traded Transaction Trade_Size_USD Status Subholding
##   <chr>   <chr>      <chr>   <chr>   <chr>         <chr>   <chr>   <chr>
## 1 NGL     ST        NGL EN~ Monda~ Sale      $15,001 - $50~ NEW    " EQUITAB~
## 2 FCX     ST        FREEPO~ Thurs~ Sale      $1,001 - $15,~ NEW    " MORGAN ~
## 3 V       ST        VISA I~ Thurs~ Purchase  $1,001 - $15,~ NEW    " TRUST O~
## 4 APPLE ~ ST        APPLE ~ Thurs~ Purchase  $360.00         NEW    " IRA ONE"
## 5 MSFT    ST        MICROS~ Thurs~ Sale      $100,001 - $2~ NEW    ""
## 6 BA      ST        BOEING~ Wedne~ Sale      $1,001 - $15,~ NEW    " BILL'S ~
## # i 11 more variables: Description <chr>, Name <chr>, Filed <chr>, Party <chr>,
## #   District <chr>, Chamber <chr>, Comments <chr>, Quiver_Upload_Time <chr>,
## #   excess_return <dbl>, State <chr>, last_modified <chr>
```

```
head(as_tibble(SP_500_annual_returns))
```

```
## # A tibble: 6 x 2
##   date      value
##   <date>    <dbl>
## 1 1928-12-31  37.9
## 2 1929-12-31 -11.9
## 3 1930-12-31 -28.5
## 4 1931-12-31 -47.1
## 5 1932-12-31 -15.2
## 6 1933-12-30  46.6
```

```
head(as_tibble(sp_500_monthly_full_data))
```

```
## # A tibble: 6 x 10
##   Date      SP500 Dividend Earnings Consumer.Price.Index Long.Interest.Rate
##   <chr>    <dbl>   <dbl>   <dbl>         <dbl>         <dbl>
## 1 1871-01-01  4.44     0.26     0.4           12.5           5.32
## 2 1871-02-01  4.5      0.26     0.4           12.8           5.32
## 3 1871-03-01  4.61     0.26     0.4           13.0           5.33
```

```
## 4 1871-04-01 4.74      0.26      0.4                12.6                5.33
## 5 1871-05-01 4.86      0.26      0.4                12.3                5.33
## 6 1871-06-01 4.82      0.26      0.4                12.1                5.34
## # i 4 more variables: Real.Price <dbl>, Real.Dividend <dbl>,
## #   Real.Earnings <dbl>, PE10 <dbl>
```

The Standard and Poor 500 annual return is sourced from Macrotrends (2025) .The annual percentage change of the S&P 500 index back to 1927. Performance is calculated as the % change from the last trading day of each year from the last trading day of the previous year.

The Standard and Poor 500 monthly return was originally compiled by Shiller (2005) and later cleaned and put into a .csv available to download on DataHub. Monthly dividend and earnings data are computed from the S&P four-quarter totals for the quarter since 1926, with linear interpolation to monthly figures. Dividend and earnings data before 1926 are from Cowles and associates, interpolated from annual data. Stock price data are monthly averages of daily closing prices through January 2000, the last month available as this book goes to press. The CPI-U (Consumer Price Index-All Urban Consumers) published by the U.S. Bureau of Labor Statistics begins in 1913; for years before 1913 I spliced to the CPI Warren and Pearson's price index, by multiplying it by the ratio of the indexes in January 1913. December 1999 and January 2000 values for the CPI-U are extrapolated.

The Congress trading data comes from Quiver Quantitative (2025). It was cleaned and put into a .csv format available for download on Kaggle by user Shabba Rank. The excess return for each recorded trade is calculated as the difference between the security's Price Change Since Trade and the SPY 's change since the trade.

These are things that are usually included in the metadata of the data set. For your project, you need to provide us with the information from your metadata that we need to understand your data set of choice.

Limitations of the data sets:

One major problem of the data sets used in this research is that its data entries are not standardized. For example, date fields are frequently saved as simple text rather than as organized date objects, which can lead to mistakes during data cleaning and make temporal analysis more difficult. The data sets also includes a lot of incomplete data lines, due to missing disclosures or incorrect filings. By distorting the representation of trading activity and possibly underestimating the timing or frequency of insider trades among Members of Congress, these inconsistencies may greatly affect the validity of the results. It is crucial to handle these data problems in order to guarantee the reliability of any insights gained from the analysis.

2.3 Describe the type of variables included

The S&P 500 annual data contains two variables: a date variable as a character object column and a value column, containing the numeric return computed as stated above for the year in the corresponding row of the date column.

The S&P 500 monthly data contains 10 variables. This includes date values and for each month the corresponding nominal and real prices, dividends and earnings, as well as Consumer price Index , PE10 and long interest rate.

The congress trading data contains 19 variables. This includes temporal and financial data for individual trades made by congress members. Relevant variables for our analysis are the trade date, the congress members' names, the trade size, the transaction type, the excess return, political party, and the transaction type. These variables are character objects or numerics. This information is compiled from disclosures by congress members required by The US government under The Stock Trading on Congressional Knowledge Act.

Part 3 - Quantifying

3.1 Data cleaning

Remove duplicate rows

Change date column from character objects to date objects

Extract numeric mean of trade size range boundaries from character objects in Trade size column

Keep only Sale transactions: only interested in realized returns of congress members

Remove NA values of trans size and excess return from data

Change Date column from character to date object

Month of October is not displayed properly month = 01, instead of month = 10 modify that and change it

Change column name to something more intuitive

Formatting date column to only years (date column entries are class == date)

Changing value column name

3.2 Generate necessary variables

Variable 1:

Calculate weighted average excess return by state over entire period:

Calculate average S&P 500 return over entire period from annual historical data:

Calculate net return size per state:

Calculate net return out-performance(under-performance) percentage:

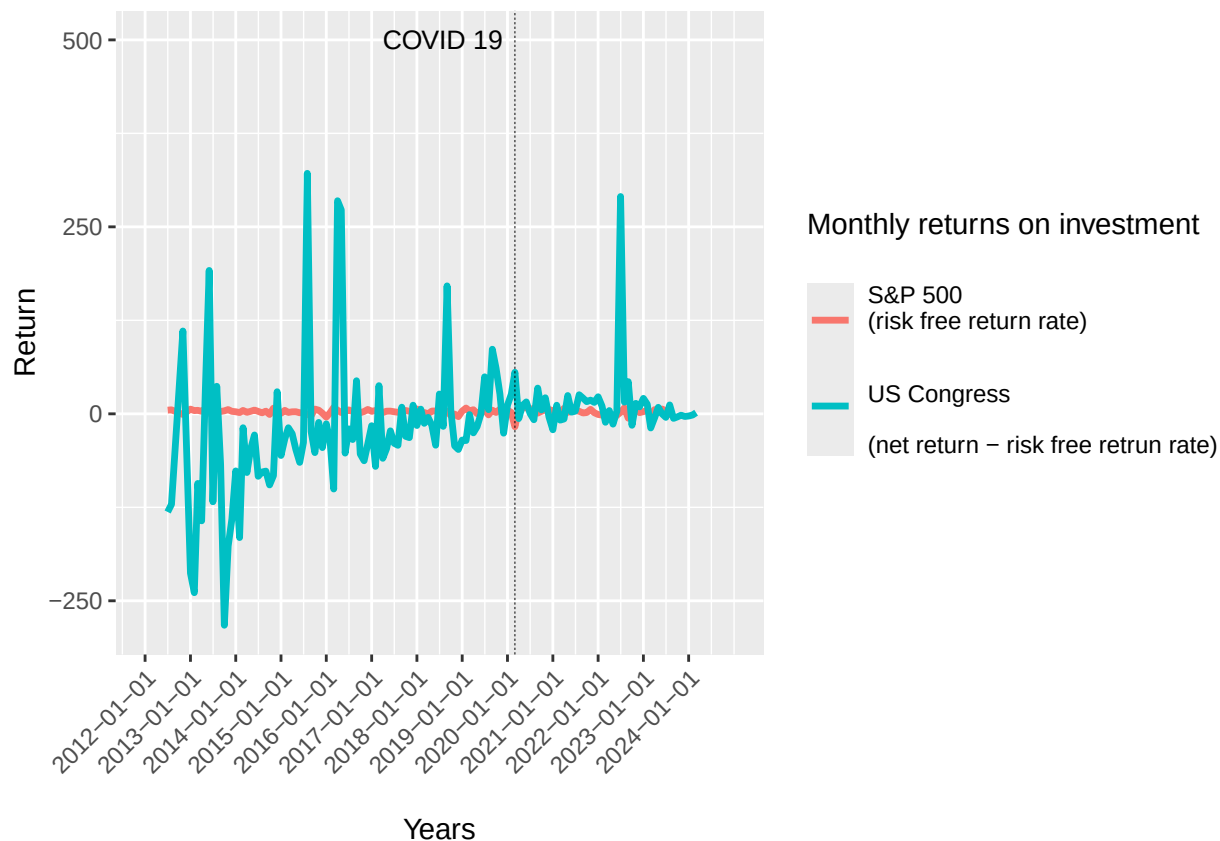
Calculate monthly return using price and dividend:

Calculate weighted average excess return for each individual congress member:

Add a Group column (categorical variable) to group congress members into positive and negative excess return groups:

3.3 Visualize temporal variation

This plots the congress members' monthly weighted average excess return (net return - risk free return) and the market's risk free monthly return rate (S&P 500 return rate) from 2012 to 2024



This shows how the return of congress members' trades react to changes in the market through time. It can reveal if congress members can anticipate changes in the market, and show the difference between the excess return and the risk free rate. This is useful to get an idea of the net return of congress members at any point in time. This can reveal possible use of insider information.

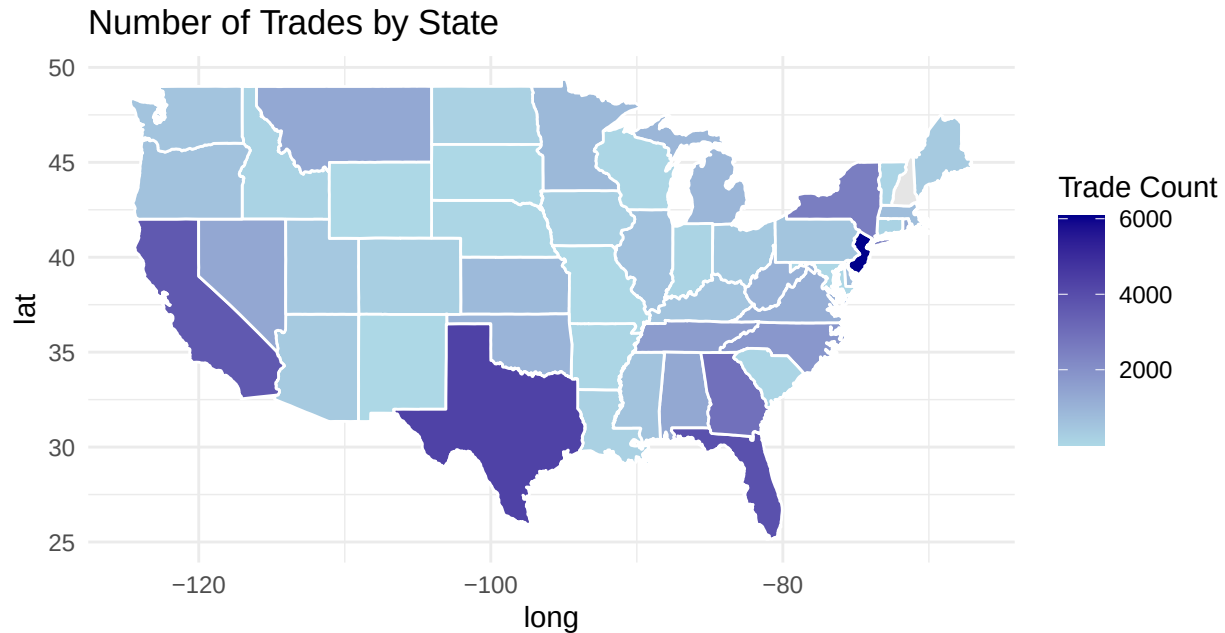
3.4 Visualize spatial variation

```
us_states <- map_data("state")
congresstrades$state_lower <- tolower(congresstrades$State)

state_trade_counts <- congresstrades %>%
  filter(!is.na(state_lower)) %>%
  group_by(state_lower) %>%
  summarise(num_trades = n())

map_data_joined <- left_join(us_states, state_trade_counts, by = c("region" = "state_lower"))

ggplot(map_data_joined, aes(x = long, y = lat, group = group, fill = num_trades)) +
  geom_polygon(color = "white") +
  coord_fixed(1.3) +
  scale_fill_gradient(low = "lightblue", high = "darkblue", na.value = "grey90") +
  labs(title = "Number of Trades by State", fill = "Trade Count") +
  theme_minimal()
```



This graph is useful for a study on insider trading in the U.S. since it clearly displays the geographic distribution of congress trades by state, which may help spot unusual or trend-related behavior. For instance:

- Trade concentrations in certain states, such as New Jersey, California, Texas, and Florida, may indicate centers of political or economic activity where insider information is more easily available or regularly acted upon.
- Outliers, like New Jersey, which has an abnormally high trading activity but a relatively small number of representatives, may point to people or groups that should be looked at more closely. However, this will take a more comprehensive and complicated study.
- Comparative analysis across states can show whether insider trading is more common in areas with more representation, economic clout, or regulatory laxity.

```
#####
## Map of net return excess percentage per state for all years 2012 --> 2024
#####
### remove alaska and hawaii
returns_by_state <- returns_by_state |>
  filter(!(State %in% c("Alaska", "Hawaii")))

##### install shapefile packages to get US states Map
devtools::install_github("ropensci/rnaturalearth")

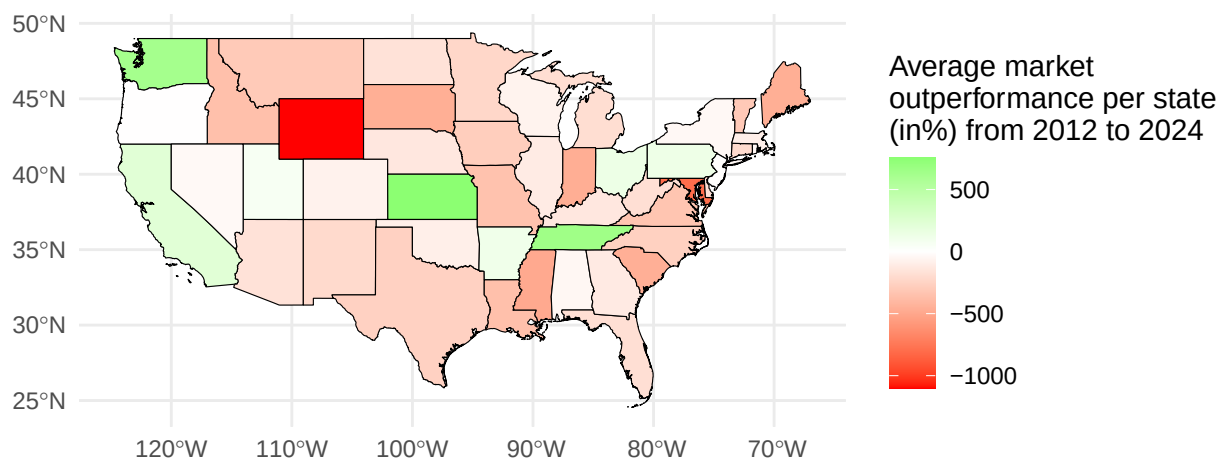
library(rnaturalearth)
## import shapefile object of US states map:

us_states_map = ne_states(country = "united states of america", returnclass = "sf" )
```

```
##### create heatmap
# merge map and returns per state data
us_states_returns_map <- us_states_map |>
  inner_join(returns_by_state, by = c("name" = "State"))

#plot merged data
ggplot(us_states_returns_map, aes(fill = state_net_return_percentage)) +
  geom_sf( color = "black", size = 0.2 ) +

scale_fill_gradient2(low = "red", mid = "white", high = "green", midpoint = 0,
  name = "Average market\noutperformance per state\n(in%) from 2012 to 2024") +
  theme_minimal()
```



Grouping the congress members by state then calculating how many times over each state out-performed (under-performed) the market rate helps visualize if abnormal returns are spatially concentrated or if the trend is more widespread. A nationwide abnormal return pattern could indicate that congress members are capable of systematically achieving abnormal returns which would point to insider trading.

3.5 Visualize sub-population variation

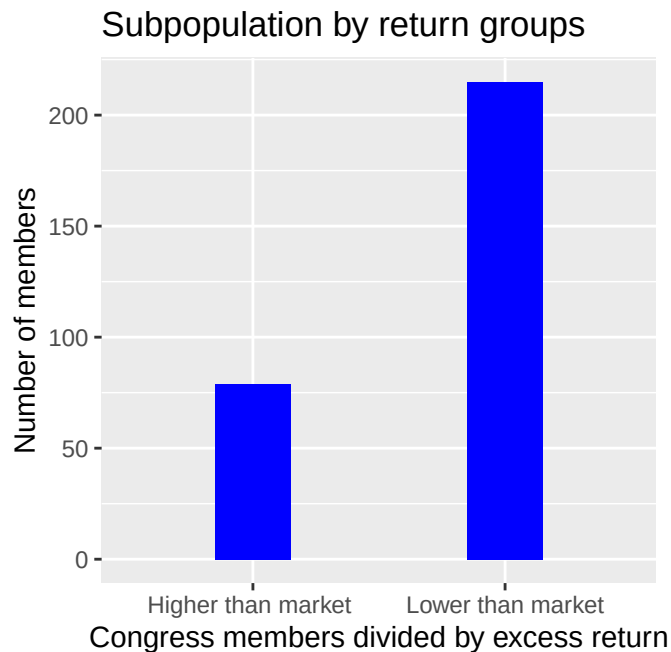
What is the number of congress members that beat the market from 2012 to 2024?

```
## Plot the number of congress members in each group
ggplot(senators_subpop_positive_excess_returns, aes(x = Group))
```

```

    ) +
  geom_bar(fill = "blue",
           width = 0.3,) +
  labs(x = "Congress members divided by excess return",
       y = "Number of members",
       title = "Subpopulation by return groups")

```



Plotting the number of congress members with higher return rates than the market and comparing it to the number of congress members with return rates lower than that of the market helps us understand whether the majority of is capable of achieving abnormal returns through insider trading. If there is a large proportion of congress members with positive excess returns that could hint at the presence of widespread access to insider information.

3.6 Event analysis

```

##### covid 19 Zoom in line plot
reshaped_monthly_returns[165:206,] |>
  filter(!is.na(Return)) |> #remove NA returns
  ggplot( aes(x=Trade_month , y=Return, group=Series, color=Series)) +
  geom_line(linewidth = 1.2) +

  scale_x_date(
    name = "\nYears",
    limits = as.Date(c("2019-06-01", "2021-01-01")) ) +

  theme(
    axis.text.x = element_text(angle = 45, hjust = 1) ) +

  scale_color_discrete(name = "Monthly returns on investment\n",
                       labels = c("sp_500_monthly_return" = "S&P 500\n(risk free return rate)\n" ,

```

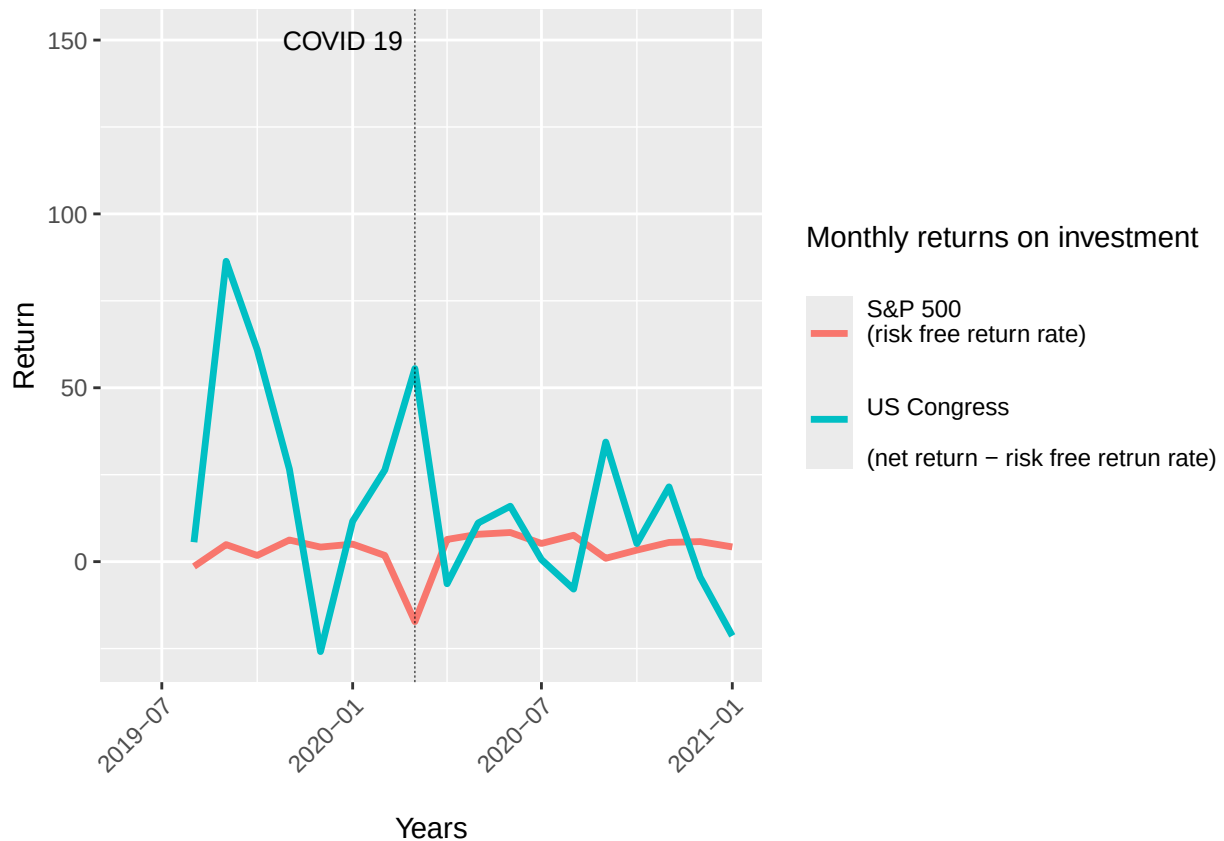


```

    "weighted_average_excess_return_monthly" =
      "\nUS Congress\n\n(net return - risk free retrun rate)")) +
geom_vline(xintercept = as.Date("2020-03-01"),
  linetype = "dashed",
  color = "black",
  size = 0.1) +

annotate("text",
  x = as.Date("2020-03-01"),
  y = 150,
  label = "COVID 19",
  hjust = 1.1,
  angle = 0,
  size = 3.5,
  color = "black")

```



This plot focuses on a time interval surrounding the Covid 19 pandemic event. It is characterized by a local minimum in the market return rate. If congress members had prior knowledge of legislation and how it would affect the market leaing up to the Covid 19 pandemic event, we would observe an upward trend in the returns on investments made by congress members.

Part 4 - Discussion

4.1 Discuss your findings

Our investigation into potential insider trading among U.S. Congress members aimed to identify patterns of abnormal returns by comparing Senate trading activity with general market performance. When calculating excess returns, we found that more than half of the Congress members generated negative excess returns, meaning their net returns were lower than those of the overall market. This finding challenges the notion that insider information is helping members achieve superior investment performance. A temporal analysis of congressional returns versus market performance from 2012 to 2024 shows relatively stable fluctuations in congressional returns, with notable spikes in 2015, 2016, and 2022. Interestingly, during the onset of the COVID-19 pandemic, congressional returns spiked while the market declined. This discrepancy raises concerns about whether members may have benefited from privileged, non-public information. When examining performance by state, Kansas, Tennessee, and Washington stood out, with members from these states consistently outperforming the market over the study period. In terms of political party, both major parties had a comparable number of trades. However, Republicans showed higher average returns until 2017, after which Democrats began to outperform in terms of excess returns. Despite these insights, limitations such as incomplete or missing disclosures hindered the depth and precision of our analysis. To fully assess the extent and impact of potential insider trading in Congress, more comprehensive and transparent data is essential. Nevertheless, our findings highlight concerns about transparency and accountability in the financial activities of lawmakers.

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/giovannivittorio/Programming_Project_Group4-4

5.2 Reference list

Abdurakhmonov, M., Snider, R. E., Ridge, J. W., & Hasija, D. (2022). Perceptions of political Self-Dealing ? An empirical investigation of market returns surrounding the disclosure of politician stock purchases. *Strategic Management Journal*. <https://sms.onlinelibrary.wiley.com/doi/10.1002/smj.3459>

Almeida, L. (2025, April 10). Trump ignites “insider trading” accusations after global tariffs U-turn. *The Guardian*; *The Guardian*. <https://www.theguardian.com/us-news/2025/apr/10/donald-trump-ignites-insider-trading-accusations-after-global-tariffs-u-turn>

Congressional Stock Trading and the STOCK Act. (n.d.). Campaign Legal Center. <https://campaignlegal.org/update/congressional-stock-trading-and-stock-act>

Himes, J. A. (2021, May 19). Text - H.R.2655 - 117th Congress (2021-2022): Insider Trading Prohibition Act. *Www.congress.gov*. <https://www.congress.gov/bill/117th-congress/house-bill/2655/text>

Data set References:

Macrotrends. (z.d.). S&P 500 Historical Annual Returns [Dataset]. <https://www.macrotrends.net/2526/sp-500-historical-annual-returns>

Shiller, R. (2005). Online Data: US Stock Markets 1871-Present and CAPE Ratio [Dataset] <http://www.econ.yale.edu/~shiller/data.htm>

Shiller R. (2024). Standard and Poor’s (S&P) 500 Index Data including Dividend, Earnings and P/E Ratio [Dataset]. DATA HUB. <https://datahub.io/core/s-and-p-500>

Quiver Quantitative (2025). Congress Trading [Dataset] <https://www.quiverquant.com/congresstrading/>

Shabba Rank. (2023). Congressional Trading (Inception to March 23) [Dataset]. Kaggle. <https://www.kaggle.com/datasets/shabbarank/congressional-trading-inception-to-march-23?select=Copy+of+congress-trading-all+%283%29.csv>